

## SMFD Assignment-4

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### Q1: European Call Option and Real Options Framing

#### (a) Implied Volatility Estimation:

Given:

- Stock Price:  $S = 38$
- Strike Price:  $K = 35$
- Time to Maturity:  $T = \frac{4}{12}$
- Risk-free Rate:  $r = 0.06$
- Market Call Option Price:  $C = 4.20$

We apply the Black-Scholes formula for a European call:

$$C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$$
$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Trying different volatilities:

| Volatility ( $\sigma$ ) | Call Price |
|-------------------------|------------|
| 0.1                     | 3.73       |
| 0.2                     | 4.12       |
| 0.3                     | 4.76       |
| 0.4                     | 5.49       |
| 0.5                     | 6.25       |

Using interpolation or numerical root-finding, the implied volatility is:

$$\sigma \approx 0.213$$

#### (b) European Put Option Price (via Put-Call Parity):

Put-Call Parity:

$$P = C + Ke^{-rT} - S$$

Substituting values:

$$P = 4.53 + 35e^{-0.06 \cdot \frac{1}{3}} - 38 \approx \boxed{0.93}$$

**(c) Real Options Interpretation:**

The launch option is equivalent to a European call with:

- Strike = \$35M (launch cost)
- Underlying = \$38M (expected revenue)
- Maturity = 4 months
- Volatility = 0.28

Using the Black-Scholes model, the value of this opportunity is \$4.20M, which is more than the intrinsic value of \$3M.

**Conclusion:** The firm should not exercise immediately but wait, as the option has positive time value. This supports deferring the investment decision.

## Q2. Pen and Paper Option Pricing (25 Marks)

### Part A: Discrete Binomial Model (European Call Option)

Given:

- Current Stock Price  $S_0 = 100$
- Strike Price  $K = 105$
- Time to Maturity  $T = 10$  days
- Option Type: European Call
- Stock moves  $\pm 1$  per day with equal probability (i.e.,  $u = +1$ ,  $d = -1$ ,  $p = 0.5$ )

**(a) Probability that the option ends in the money after 10 days:**

Let  $X$  be the number of up-moves in 10 days. Then the final stock price is:

$$S_{10} = 100 + (2X - 10)$$

We want  $S_{10} > 105 \Rightarrow 100 + 2X - 10 > 105 \Rightarrow X > 7.5 \Rightarrow X \geq 8$   
So, the probability is:

$$P(X \geq 8) = P(X = 8) + P(X = 9) + P(X = 10)$$

Using the binomial distribution:

$$P(X = k) = \binom{10}{k} \cdot 0.5^{10}$$

$$P = \frac{1}{1024} \left( \binom{10}{8} + \binom{10}{9} + \binom{10}{10} \right) = \frac{1}{1024} (45 + 10 + 1) = \frac{56}{1024} = \frac{7}{128} \approx 0.0547$$

**(b) Expected payoff of the option:**

Payoff for final price  $S_{10} > 105$  is  $S_{10} - 105$ , otherwise 0.

We compute:

$$\begin{aligned} E[\text{Payoff}] &= \sum_{k=8}^{10} \left[ \binom{10}{k} \cdot 0.5^{10} \cdot (S_{10} - 105) \right] \\ &= \frac{1}{1024} \left[ \binom{10}{8} \cdot (106 - 105) + \binom{10}{9} \cdot (108 - 105) + \binom{10}{10} \cdot (110 - 105) \right] \\ &= \frac{1}{1024} [45 \cdot 1 + 10 \cdot 3 + 1 \cdot 5] = \frac{1}{1024} (45 + 30 + 5) = \frac{80}{1024} = \frac{5}{64} \approx 0.0781 \end{aligned}$$

**(c) Fair Value of the Option (ignoring discounting):**

Since this is a risk-neutral world and we ignore discounting, the fair value of the option is the expected payoff:

|                                                   |
|---------------------------------------------------|
| $\text{Fair Value} = \frac{5}{64} \approx 0.0781$ |
|---------------------------------------------------|

## Q2 Part B: Continuous Normal Distribution Model

We assume that the stock's daily return follows a normal distribution with:

- Mean = 0
- Standard deviation  $\sigma$  such that the expected absolute move is \$1 per day

**(a) Finding  $\sigma$  from expected absolute move**

We use the identity:

$$\mathbb{E}[|X|] = \sigma \sqrt{\frac{2}{\pi}}$$

Solving for  $\sigma$ :

$$\sigma = \frac{1}{\sqrt{\frac{2}{\pi}}} \approx 1.2533$$

To scale this to 10 days (assuming independent daily returns), we use:

$$\sigma_{10} = \sigma \sqrt{10} \approx 1.2533 \times \sqrt{10} \approx 3.963$$

### (b) Expected payoff of the option

Let  $S_T$  be the terminal stock price after 10 days, which is normally distributed with:

$$S_T \sim \mathcal{N}(100, \sigma_{10}^2)$$

Then the expected payoff of the European call is:

$$\mathbb{E}[\max(S_T - 105, 0)] = \int_{105}^{\infty} (S - 105) f_{S_T}(S) dS$$

### (c) Numerical Evaluation via Python

```
import numpy as np
from scipy.stats import norm
from scipy.integrate import quad

S0 = 100
K = 105
sigma = 1.2533 * np.sqrt(10)
mu = S0
std = sigma

# PDF of normal distribution
f = lambda S: norm.pdf(S, loc=mu, scale=std)

# Integrand: (S - K) * f(S)
integrand = lambda S: (S - K) * f(S)

# Compute the integral from K to infinity
expected_payoff, _ = quad(integrand, K, np.inf)

print("Expected Payoff of Call Option: $", round(
    expected_payoff, 4))
```

This evaluates the fair value of the call option under the normal distribution assumption.

## Q2 Part C: Uniform Distribution Model

### (a) Finding $[a, b]$ such that $\mathbb{E}[|X|] = 1$

Let  $X \sim \mathcal{U}(-a, a)$  with symmetric support. Then:

$$\mathbb{E}[|X|] = \frac{a}{2} \Rightarrow a = 2 \Rightarrow X \sim \mathcal{U}(-2, 2)$$

So the daily change is from the uniform interval  $[-2, 2]$ .

## (b) Comparison of Models

- Binomial: Discrete distribution; prices move by  $\pm 1$
- Normal: Continuous distribution centered at 0 with standard deviation  $\approx 1.25$
- Uniform: Equal probability of movement in  $[-2, 2]$  range

The final 10-day price distributions will differ in smoothness and tails.

## (c) Monte Carlo Simulation Method

- Simulate  $N$  paths of 10-day returns using daily uniform random draws in  $[-2, 2]$
- Compute the terminal stock price  $S_T = S_0 + \sum_{i=1}^{10} X_i$
- Calculate the payoff:  $\max(S_T - K, 0)$
- Average over all simulations for expected payoff

```
np.random.seed(0)
S0 = 100
K = 105
days = 10
n_sim = 100000

# Simulate 100000 paths with uniform steps in [-2, 2]
daily_returns = np.random.uniform(-2, 2, size=(n_sim, days))
final_prices = S0 + daily_returns.sum(axis=1)

payoffs = np.maximum(final_prices - K, 0)
expected_payoff = np.mean(payoffs)

print("Monte Carlo Estimated Option Price: $", round(
    expected_payoff, 4))
```

This simulation gives us the fair value of the option under the uniform model.